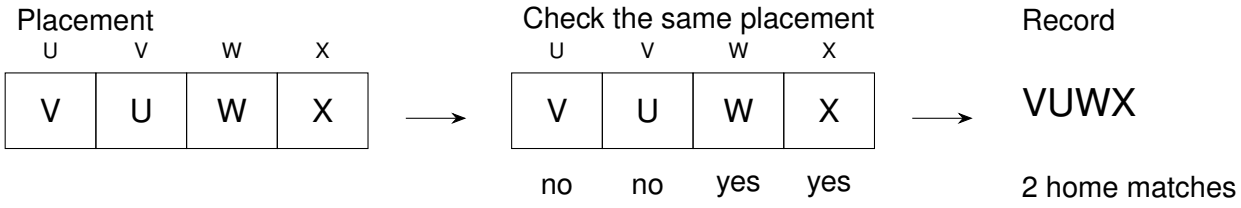
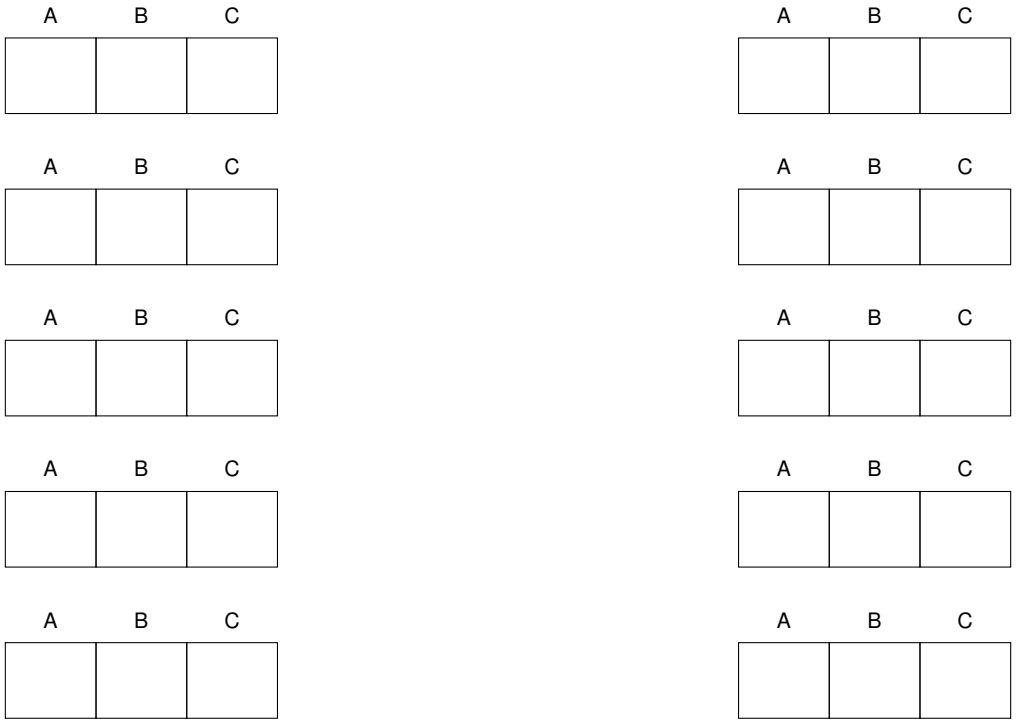
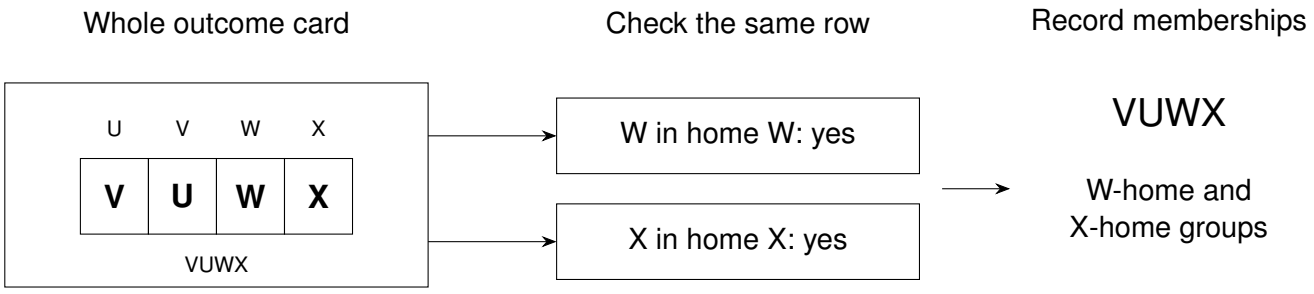
Use a card set and the home strip with the same letters. Use each card once, with one card in every home. Keep the home labels fixed. A *home match* means the card and its home have the same letter. Check a complete row before moving any card. Record the cards from left to right. A row with no home matches is *home-free*.



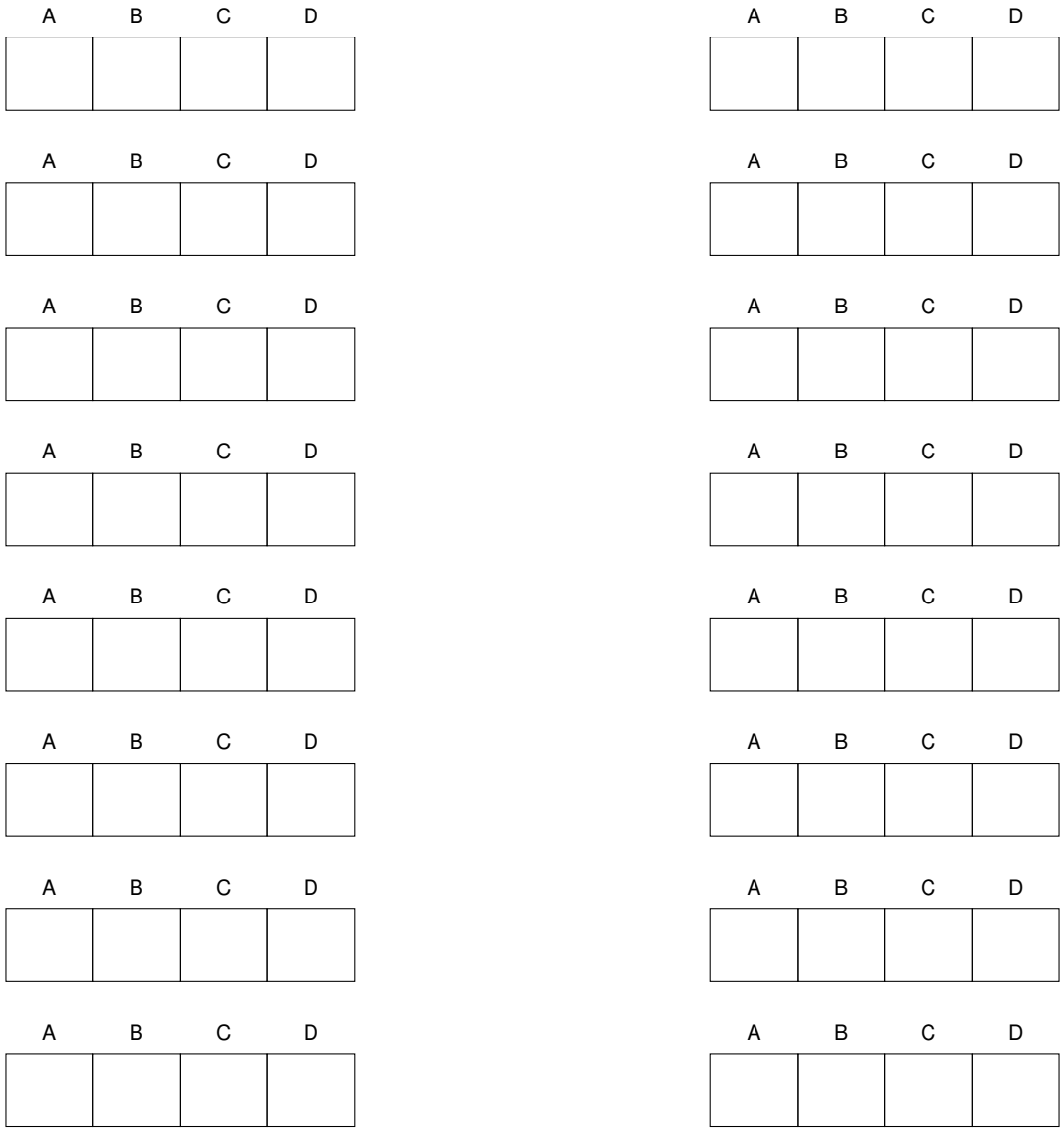
Problem 1: Find every row of A, B, C with no home matches. Can a row of these three cards have exactly two home matches?



One outcome card is one whole placement.



Problem 2: Use the six A–C outcome cards. Make one group with A in home A and another with B in home B. A card may belong to both groups. How many rows are outside both groups? C may stay home.

A student counts $6 - 2 - 2 = 2$. Repair this count and show what went wrong.

Problem 3: Find every A–D row with no home matches. Make a catalog your table can check for repeated or missing rows.